

Linear search in C

Linear search in C to find whether a number is present in an array. If it's present, then at what location it occurs. It is also known as a sequential search. It is straightforward and works as follows: we compare each element with the element to search until we find it or the list ends. Linear search for [multiple occurrences](#) and using a [function](#).

Linear search program in C

```
#include <stdio.h>

int main()
{
    int array[100], search, c, n;

    printf("Enter number of elements in array\n");
    scanf("%d", &n);

    printf("Enter %d integer(s)\n", n);

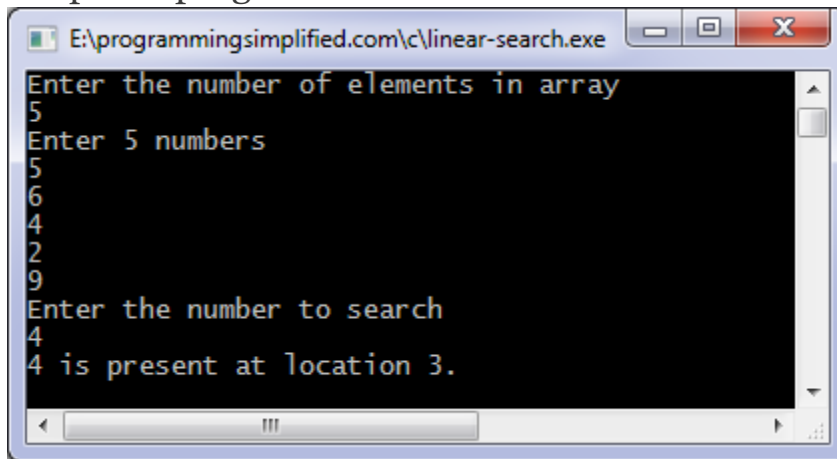
    for (c = 0; c < n; c++)
        scanf("%d", &array[c]);

    printf("Enter a number to search\n");
    scanf("%d", &search);

    for (c = 0; c < n; c++)
    {
        if (array[c] == search)      /* If required element
is found */
        {
            printf("%d is present at location
%d.\n", search, c+1);
            break;
        }
    }
    if (c == n)
        printf("%d isn't present in the
array.\n", search);

    return 0;
}
```

Output of program:



```
E:\programmingsimplified.com\c\linear-search.exe
Enter the number of elements in array
5
Enter 5 numbers
5
6
4
2
9
Enter the number to search
4
4 is present at location 3.
```

The image shows a screenshot of a Windows command prompt window. The title bar at the top reads "E:\programmingsimplified.com\c\linear-search.exe". The window contains the following text: "Enter the number of elements in array", "5", "Enter 5 numbers", "5", "6", "4", "2", "9", "Enter the number to search", "4", and "4 is present at location 3.". The text is displayed in a monospaced font on a black background. The window has standard Windows window controls (minimize, maximize, close) in the top right corner.